Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.

S = {1, 2, 3………, 400}. For how many non-empty subsets of S is the product of the elements of the subset equal to an even number?

a) $2^{300}(2^{100} - 1)$

b) $2^{100}(2^{300} - 1)$

c) $2^{200}(2^{200} - 1)$

d) $2^{200} - 1$

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.

What is the largest number which leaves remainders of 4, 6 and 2 when it divides 460, 690 and 990 respectively?

Q3. DIRECTIONS *for question 3:* Select the correct alternative from the given choices.

In a class, the number of students who passed in at most 0, 1, 2, 3, 4 and 5 subjects is 3, 7, 9, 14, 15 and 22 respectively. If no student passed in more than 5 subjects, find the number of students who passed in at least 3 subjects.

- a) **8**
- b) **13**
- c) **14**
- d) **15**

Q4. DIRECTIONS *for questions 4 and 5:* Type in your answer in the input box provided below the question.

If the cost of painting the walls of a cuboidal room of length 15 ft and breadth 12 ft, at ₹25 per sq. ft, is ₹10,800, find the height (in ft) of the room (Ignoring any doors and windows in the room).

Q5. DIRECTIONS *for questions 4 and 5:* Type in your answer in the input box provided below the question.

The visibility on a certain road is limited to a distance of 80 m due to fog. A car travelling on that road crossed a cart moving in the same direction. If the speed of the cart and the car are 8 km/hr and 44 km/hr respectively, then for how long (in seconds) will the car be visible to the driver of the cart after it overtook the cart?

Q6. DIRECTIONS *for questions 6 to 9:* Select the correct alternative from the given choices.

There are two series of numbers in geometric progression, each having the same number of terms, such that the fourth term and the sixth term of the first series are equal to the second term and the third term of the second series respectively. If the last term of the second series is 64 times the last term of the first series and the fifth term of the first series is eight times the second term of the first series, then find the number of terms present in each of the series.

- a) **8**
- b) **6**
- c) **10**
- d) **9**

Q7. **DIRECTIONS** *for questions 6 to 9:* Select the correct alternative from the given choices.
A is five years older than B and five years younger than C. If the average of the ages of A, B and C is 19 years, what is the age of B?

a) **14 years** ✓ **Your answer is correct**

b) **9 years**

c) **19 years**

d) **13 years**

Q8. DIRECTIONS *for questions 6 to 9:* Select the correct alternative from the given choices.

If the cost price, the selling price and the marked price of an article are in arithmetic progression and it is known that a profit was registered by selling the article, then which of the following statements is true?

- a) **The profit percentage was less than the discount percentage.**
- b) **The profit percentage was equal to the discount percentage.**
- c) **The mark up percentage was double the profit percentage.**
- d) **The profit percentage when calculated on the selling price was more than the actual profit percentage (i.e., when calculated on the cost price).**

Q9.

DIRECTIONS *for questions 6 to 9:* Select the correct alternative from the given choices. $\frac{1}{1!}\times\frac{2}{2!}\times\frac{3}{3!}\times\frac{4}{4!}\ldots\ldots\frac{n}{n!}$ =

a) $\frac{1!+2!+3!\ldots\ldots+(n-1)!}{n!}$

b) $\frac{1}{2^{n-1}3^{n-2}4^{n-3}\ldots\ldots(n-1)^{2}}$

c) $\frac{1}{2^{n-2}3^{n-3}4^{n-4}\ldots\ldots(n-1)^{1}}$

d) $\frac{n}{n!\,(n-1)!}$

Q10. DIRECTIONS *for questions 10 and 11:* Type in your answer in the input box provided below the question.

N is a three-digit number in the number system to the base 6. If the order of the digits is reversed and the number is then considered in base 11, its value becomes four times the value of the original number in base 6. How many values are possible for N?

Q11. DIRECTIONS *for questions 10 and 11:* Type in your answer in the input box provided below the question.

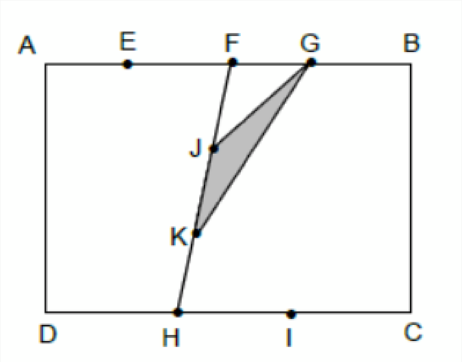
If $A = 1 + 2 + 2^2 + 2^3 + \ldots.+ 2^n$ and $B = 1 + 4 + 4^2 + 4^3 + \ldots\ldots.+ 4^m$, and $A = 3B$, find the value of $n - 2m$.

Q12. DIRECTIONS *for question 12 and 13:* Select the correct alternative from the given choices.

If $x \in R - \{-1, -3\}$, and $y = \frac{x+2}{(x+1)(x+3)}$, find the range of y.

- a) **[0, 3]**
- b) **[–2, 5]**
- c) $\left[\frac{1}{3}, \infty\right)$
- d) $(-\infty, \infty)$

Q13. DIRECTIONS *for question 12 and 13:* Select the correct alternative from the given choices.



In the above figure, given that FJ = JK = KH, AE = EF = FG = GB, DH = HI = IC and AB = 2AD = 3DH = 4FG, find the ratio of the area of the shaded region to that of the rectangle ABCD.

a) **3 : 16**

b) **1 : 18**

c) **3 : 32**

d) **1 : 24**

Q14. DIRECTIONS *for question 14:* Type in your answer in the input box provided below the question.

How many two-digit numbers have their square as 1 more than a multiple of 24?

Q15. DIRECTIONS *for questions 15 to 17:* Select the correct alternative from the given choices.

If $13^{(\log_x 872)} = 5^{(\log_{25} 961)} + 7^{(\log_{\sqrt{7}} 29)}$, then the value of *x* is

a) $\sqrt{13}$.

b)

$13\sqrt{3}$.

c) **169.**

d) **13.**

Q16. **DIRECTIONS** *for questions 15 to 17:* Select the correct alternative from the given choices.
A permutation is an ordered arrangement of two or more objects. The interchange of any two adjacent objects in a permutation is called a *transposition*. Thus, for three objects a, b and c, the permutation *acb* can be obtained from the permutation *abc* using one *transposition*.

How many *transpositions* are needed to completely reverse the permutation *abcdef*, to obtain the permutation *fedcba*?

- a) **15**
- b) **21**
- c) **6**
- d) **18**

Q17. DIRECTIONS *for questions 15 to 17:* Select the correct alternative from the given choices.

If an article is marked up by 60% above its cost price and then sold at $\frac{3}{4}^{th}$ of its marked price, find the percentage of profit made on the article.

- a) **20%** ✓ Your answer is correct
- b) **25%**
- c) **30%**
- d) **40%**

Q18. DIRECTIONS *for question 18:* Type in your answer in the input box provided below the question.

Two friends, Turbo and Black Shadow, simultaneously left two places A and B respectively. Turbo left from A, for B, and Black Shadow left from B, for A. However, every time that either of them arrives at his destination, he immediately turns back to reach his starting point and then keeps on travelling in that manner between A and B. If the distance between A and B is 7 km and the speeds of Turbo and Black Shadow are 70 kmph and 20 kmph respectively, how many times do they meet in the first seven hours after they start?

Q19. DIRECTIONS *for questions 19 to 21:* Select the correct alternative from the given choices.
Over a period of 11 days, a vegetable vendor visited at most ten villages for selling vegetables. On each day he visited exactly one village, but he did not revisit any village within two days of visiting it. In how many ways could he have visited the villages in the period of 11 days?

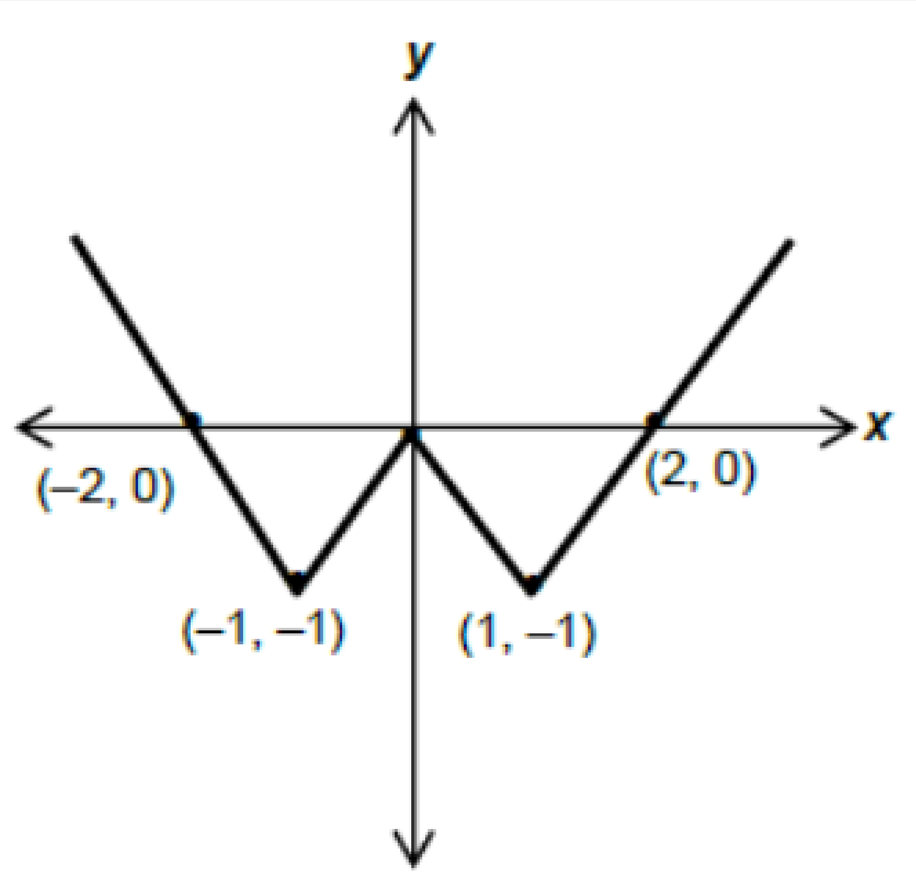
- a) $\mathbf{10(9)^{10}}$
- b) $\mathbf{90(8)^{9}}$
- c) $\mathbf{10(8)^{10}}$
- d) $\mathbf{720(7)^{8}}$

Q20. **DIRECTIONS** *for questions 19 to 21:* Select the correct alternative from the given choices.

The income of A is ₹15,000 and it is equal to the expenditure of B. If the ratio of the savings of A to the savings of B is 2 : 1, which of the following statements is definitely true?

- a) **The combined income of A and B is more than ₹45,000.**
- b) **The combined expenditure of A and B is not less than ₹20,000.**
- c) **A's expenditure added to twice of B's income is equal to ₹45,000.** ✓Your answer is correct
- d) **B's expenditure added to twice of A's income is equal to ₹30,000.**

Q21. DIRECTIONS *for questions 19 to 21:* Select the correct alternative from the given choices. Which of the following functions best describes the graph given below?



- a) $y = | |x| + 1| - 2$
- b) $y = | |x| - 2| - 1$
- c) $y = | |x| - 1| - 1$
- d) $y = | |x - 1| - 1|$

Q22. **DIRECTIONS** *for question 22:* Type in your answer in the input box provided below the question.

A cuboid of dimensions 4 cm × 5 cm × 6 cm is constructed from identical cubes, each of side 1 cm. The cuboid is then placed on a table, resting on one of its 6 cm × 4 cm face. Now, two vertical cuts are made (down the height), one cut along each of the two diagonals of the top face. What is the total number of unit cubes which are not affected by the cuts?